Secure CLI Password Manager

Cryptographic Architecture & Implementation Report

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Python & Information Security Project

01. EXECUTIVE SUMMARY

This technical report documents the design, architecture, and cryptographic specifications of a zero-knowledge local Command-Line Interface (CLI) Password Manager engineered by **Shaik Mahummed Arshad**. The system eliminates vulnerabilities associated with plain-text credential persistence by implementing a multi-tier cryptographic pipeline. Utilizing **Argon2id** for user authentication, **PBKDF2HMAC-SHA256** for key derivation, and **Fernet (AES-128-CBC with HMAC-SHA256)** for payload encryption at rest, the manager guarantees strong confidentiality and tamper resistance backed by an embedded SQLite database.

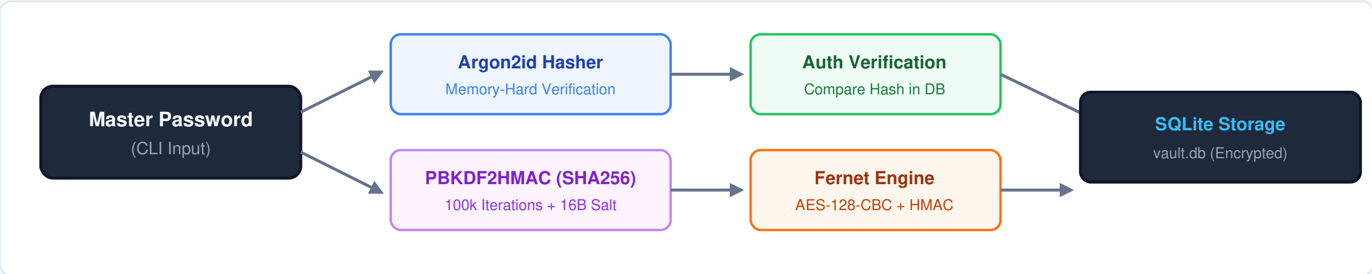
Problem Statement

Conventional password storage using unencrypted files or unsalted algorithms (MD5, SHA-1, standard SHA-256) exposes credentials to instant exfiltration, rainbow table lookups, and rapid GPU brute-forcing upon unauthorized system access.

Core Security Objectives

- **Zero-Knowledge Principle:** Raw master keys & passwords never touch non-volatile storage.
- **Hardware Resistance:** Memory-hard derivation counters ASIC/GPU parallel attacks.
- **Cryptographic Integrity:** Authenticated symmetric cipher prevents bit-flipping.

02. CRYPTOGRAPHIC ARCHITECTURE & WORKFLOW



03. DATABASE SCHEMA & STORAGE SPECIFICATIONS

TABLE	COLUMN NAME	DATA TYPE	DESCRIPTION & SECURITY ROLE
`config`	`master_hash`	`TEXT`	Argon2id hash string storing salt, cost, and digest for login verification.
`config`	`salt`	`BLOB`	Cryptographically secure 16-byte random salt for encryption key derivation.
`secrets`	`site`	`TEXT`	Target domain identifier, service name, or website URL.
`secrets`	`username`	`TEXT`	Associated user identifier, email handle, or account username.
`secrets`	`encrypted_password`	`TEXT`	Authenticated Fernet ciphertext token (AES-128-CBC + HMAC-SHA256).

04. SECURITY ANALYSIS & THREAT MITIGATION

Brute-Force & ASICs	Argon2id combines data-dependent and data-independent memory access, significantly penalizing custom ASIC/GPU cracking efficiency compared to raw SHA-256.

Rainbow Tables	Unique 16-byte random salts per database instance guarantee that identical passwords result in completely distinct hashes and derived keys.
Cipher Tampering	Fernet's built-in HMAC-SHA256 signing ensures that any unauthorized byte manipulation within SQLite triggers an immediate decryption validation failure.

05. EVALUATION & FUTURE EXTENSIONS

Completed Milestones

- Full CLI workflow (Setup, Add, View, Validate).
- Zero plain-text credential persistence.
- Robust exception handling on auth mismatch.

Planned Enhancements

- Automated clipboard autofill with 15s clear timer.
- Integrated cryptographically secure password generator.
- TOTP / Two-Factor Authentication module support.

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Status: Completed & Verified